DO NOW — RECALL

Floating point, binary and Boolean mathematics

Name: _____ Class: _____ Date: _____

1. What is normalisation's purpose?

2. What is first-two-bit rule for normalised two's-complement mantissa?

3. State De Morgan: NOT(A AND B).

4. Simplify (A∧B)∨(¬A∧B) with working.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Standardised mantissa maximising precision.

2. Model answer

The first two bits differ.

3. Model answer

NOT A OR NOT B.

4. Model answer

$B \wedge (A \vee \neg A) = B$.

By the end, you can…

• Normalising signed binary mantissas shifts bits and compensates exponent to preserve the same value.

• For negative integer conversion, write exactly eight positive bits, invert each, then add one.

• For algebraic simplification, show each Boolean law used rather than only the final expression.

Vocabulary — know these terms

mantissa — Significand representing precision.

exponent — Power-of-two scale.

two's complement — Representation for signed integers.

normalisation — Shift/scaling preserving represented value.

NEW LEARNING

Read, annotate and explain

Understand

Normalising signed binary mantissas shifts bits and compensates exponent to preserve the same value.

Apply

For negative integer conversion, write exactly eight positive bits, invert each, then add one.

Evaluate

For algebraic simplification, show each Boolean law used rather than only the final expression.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Normalise $0.0101_2 \times 2^4$.

Predict first

Left-shift mantissa once and lower exponent by one.

Explain and check

Explain the reason for the result.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Normalise $0.00101_2 \times 2^5$ and express its decimal value. [6]

Question 2

Convert −23 to 8-bit two's complement, showing every step. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Simplify A∨(¬A∧B) using identities; verify for A0,B1. [6]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

Normalisation shifts mantissa but never changes exponent.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.